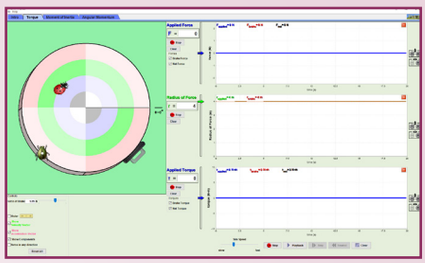
ICT CORNER

Moment of inertia

Which is harder to rotate:
Circular ring or Circular disc?

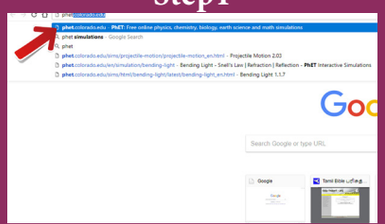


STEPS:

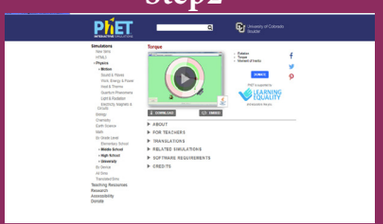
- Open the browser and type the given URL to open the PhET simulation on Torque. Click the picture link or the download button. Once downloaded, click ok to open the java applet.
- Set platform mass 0.1 kg, Outer radius 4 m. (Keep inner radius = 0). Now it is a circular disc. Click the button 'go' to get the value of moment of inertia.
- Adjust the values of mass and radius and then observe how moment of inertia changes in the middle graph.
- Keep the inner and outer radius same (say R = r = 4 m.) and mass 0.1 kg. Now it becomes circular ring. Click the button 'go' to start the calculation.
- Observe the moment of inertia from the middle graph. Compare the moment of inertia of a circular disc and circular ring with same mass and radius.

Hint: If moment of inertia is relatively large, it is very difficult to accelerate in angular direction.

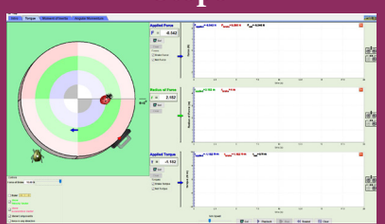
Step1



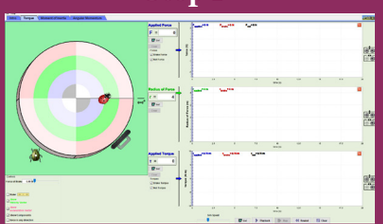
Step2



Step3



Step4



PhET simulation's URL:

https://phet.colorado.edu/en/simulation/torque

* Pictures are indicative only.

*If browser requires, allow Flash Player or Java Script to load the page.

UNIT-5(XI-Physics_Vol-1).indd 265 06-01-2022 17:48:22